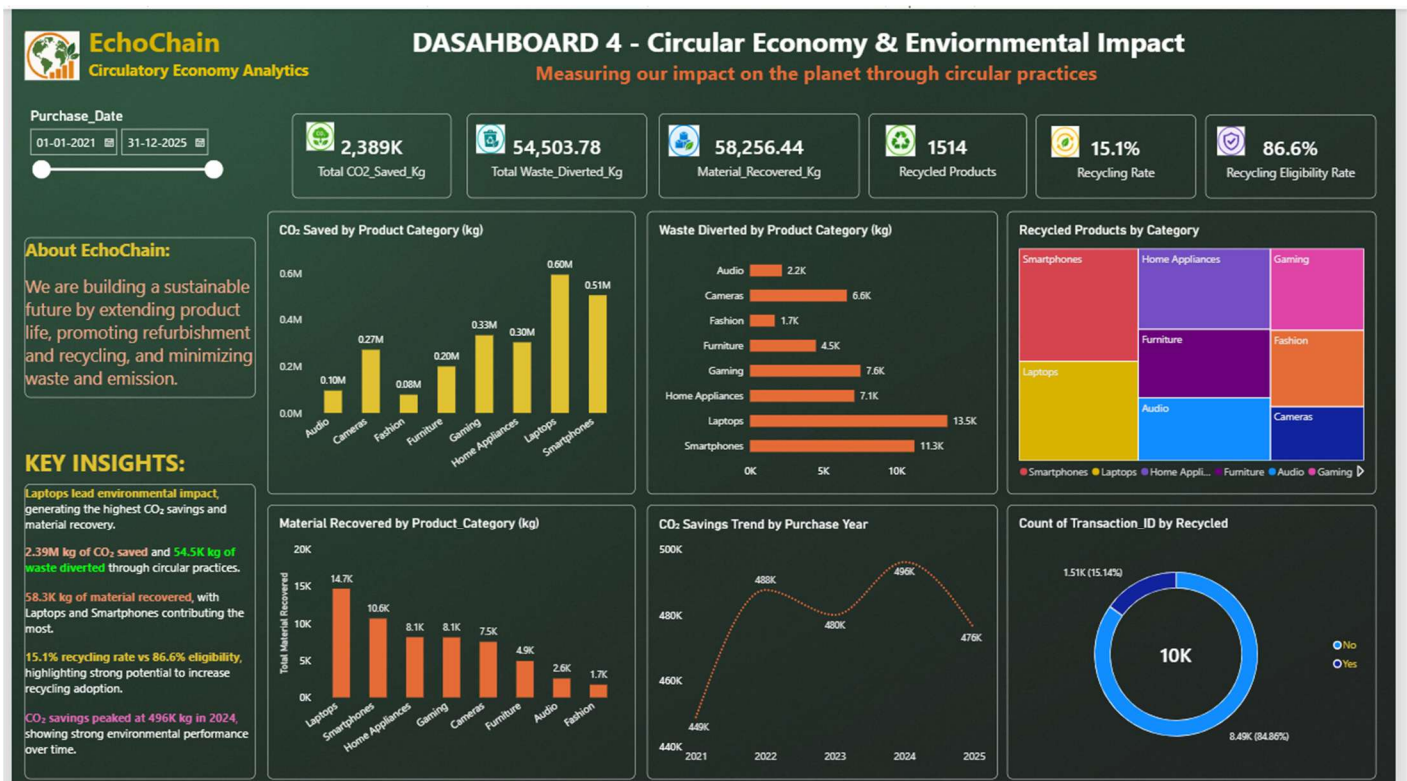


Dashboard 4 – Circular Economy & Environmental Impact:

This dashboard provides an overview of EchoChain's circular economy performance and environmental impact across different product categories. It helps evaluate CO₂ savings, waste diversion, material recovery, and recycling performance, giving a clear picture of how circular practices extend product life and reduce environmental footprint.

The dashboard combines environmental and operational metrics to identify **high-impact product categories, waste diversion leaders, recycling performance, and yearly CO₂ savings trends**. The interactive purchase-date filter also lets users analyse these insights for a selected period.

Dashboard 4 – Image Overview:



Key outcomes from the dashboard:

- **Total Environmental Impact:** EchoChain records approximately **2.39 million kg of CO₂ saved, 54,503.78 kg of waste diverted, and 58,256.44 kg of material recovered** across the analyzed lifecycle data.
- **Laptops Lead Environmental Impact:** Laptops contribute the highest CO₂ savings at approximately **0.60 million kg**, followed by Smartphones at **0.51 million kg**.
- **Waste Diversion Leaders:** Laptops have the highest waste diversion at approximately **13.5K kg**, followed by Smartphones at **11.3K kg** and Gaming products at **7.6K kg**.
- **Material Recovery:** Laptops lead material recovery with **14.7K kg**, followed by Smartphones with **10.6K kg**. Home Appliances and Gaming both contribute approximately **8.1K kg**.
- **Recycling Performance:** **1,514 products** have been recycled, representing a **15.1% recycling rate**, while **86.6% of products are eligible for recycling**.
- **Recycling Status:** The majority of transactions, approximately **84.86%**, are currently classified as not recycled, while **15.14%** have been recycled. This highlights an opportunity to increase recycling conversion among eligible products.
- **CO₂ Savings Trend:** Annual CO₂ savings remained relatively strong throughout 2021–2025, increasing from approximately **449K kg in 2021** to a peak of **496K kg in 2024**, before declining slightly to approximately **476K kg in 2025**.
- **Category-Level Opportunity:** **Fashion and Audio** show the lowest environmental contributions across CO₂ savings, waste diversion, and material recovery, indicating potential areas for improving circularity practices.